

TEAM ACE PRESENTATION

SMART DRAIN & HEATWAVE RESPONSE SYSTEM

An integrated climate response platform that analyzes real-time weather API and environmental sensor data to proactively manage both **heatwave droughts** (automated micro-irrigation) and **torrential rain** (pre-storm smart drainage) — using a single storage tank.

SCALE

1:1 Physical Prototype

TECH STACK

**ESP32 · MQTT · Raspberry Pi
4**

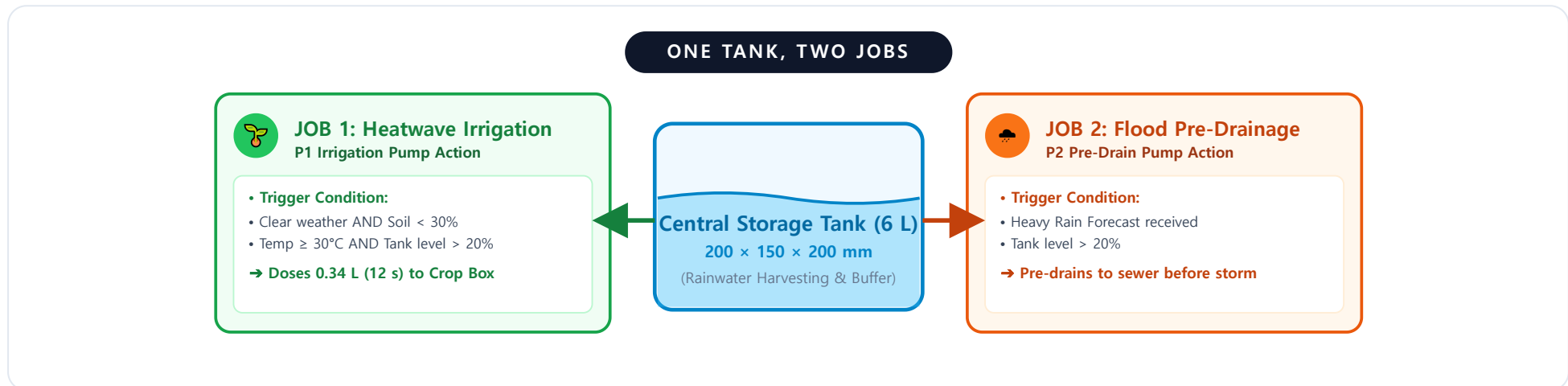
DATA SOURCE

KMA Forecast Open API

DEVELOPMENT STATUS

Logic Verified & Complete

One Tank, Two Jobs: Store Rainwater, Empty Before Storm



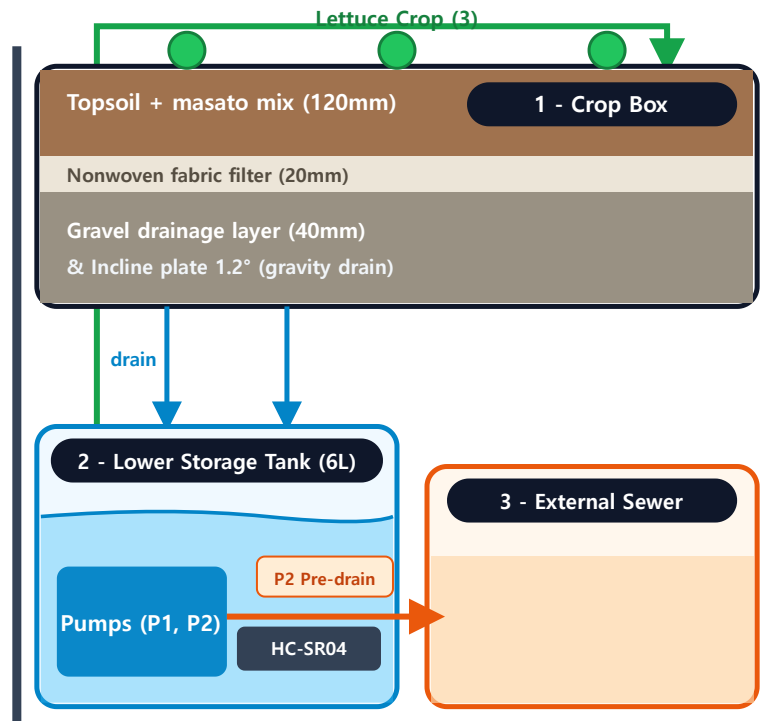
⚡ Two Hazards Arriving Together

- **Urban Heat Island & Drought:** Summer heat stresses crops and spikes urban discomfort index.
- **Torrential Rain:** Flash flooding overwhelms standard storm sewers in minutes.
- **Compound Events:** Heatwaves and downpours happen in rapid succession within the same week.

💡 Bi-Directional Integrated Solution

Responding to heavy rain alone is no longer enough. **Team ACE's SDS Platform manages both hazards using a single 6 L storage tank** — automatically toggling between **pre-storm capacity clearing** and **dry-period micro-dosing irrigation** based on 5-second sensor & forecast evaluations.

2D Smart Irrigation & Drainage System Mechanism



1. Top Crop Box Container

- Dimensions: **400 × 300 × 249 mm**.
- **3-layer soil profile:** Topsoil + masato (120mm), Nonwoven fabric filter (20mm), Gravel drainage layer (40mm).
- Houses 3 lettuce crops with 1.2° incline plate for gravity drainage.

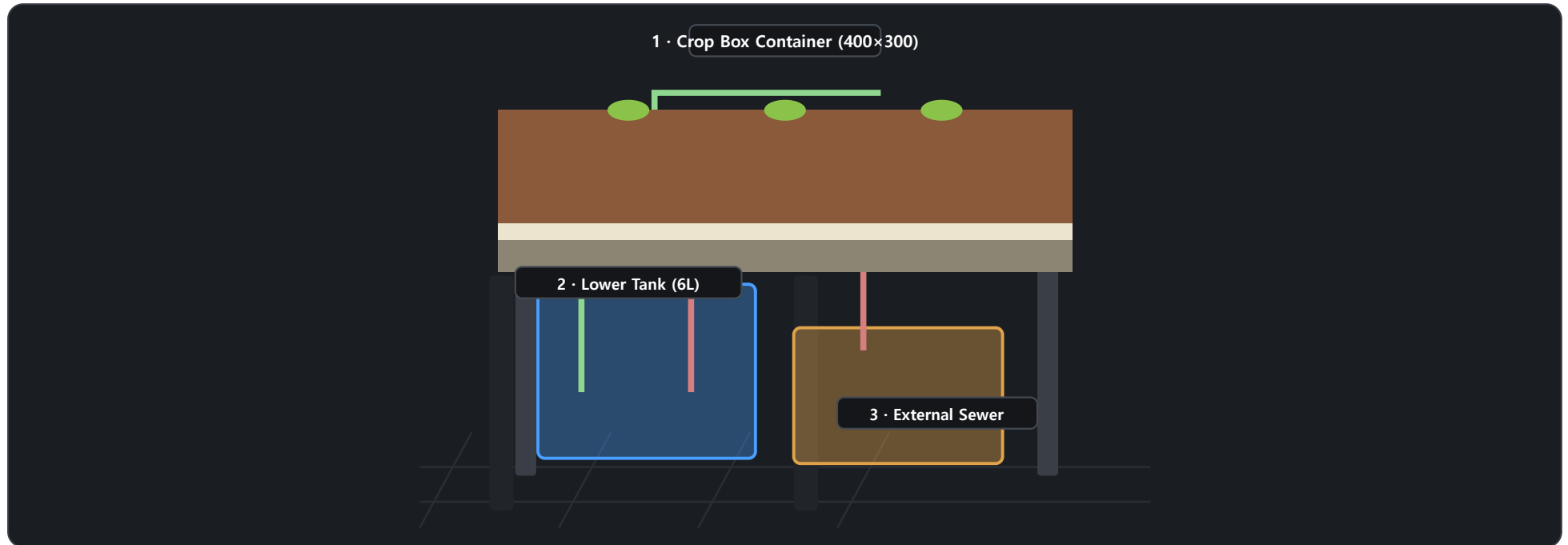
2. Lower Storage Tank

- Dimensions: **200 × 150 × 200 mm (6 L Capacity)**.
- Real-time volume tracking via **HC-SR04 ultrasonic sensor**.
- Dual submersible pumps (P1 & P2) for bi-directional water routing.

3. External Sewage & Pump Circuits

- **External Sewage (150 × 200 × 100 mm)** for pre-storm waste isolation.
- **P1 (Irrigation):** Lower Tank → Top Crop Box drip nozzle.
- **P2 (Pre-drainage):** Lower Tank → External storm sewer.

3D CAD Digital Twin Model Display



* Three.js WebGL 3D CAD Model Digital Twin Rendering (1:1 Physical Scale)

① 1:1 Scale Precision 3D Model

Three.js WebGL engine visualizes **400×300×249mm crop box** and storage tanks with exact 1:1 physical CAD scale.

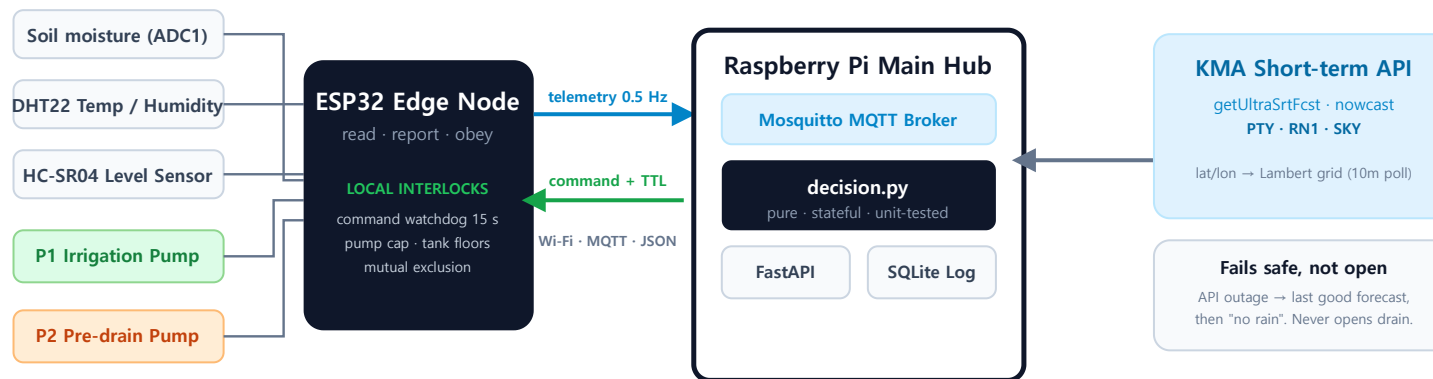
② Dual Flow Route Verification

Pre-verifies mechanical clearance and zero-interference between top irrigation pipe and external drainage outlet.

③ STL / OBJ CAD Data Export

Export real-time 3D model geometry directly into **STL / OBJ format** for 3D printing and physical manufacturing.

Dumb Edge Node, Smart Hub Architecture



Telemetry — ESP32 → Pi, every 2 s

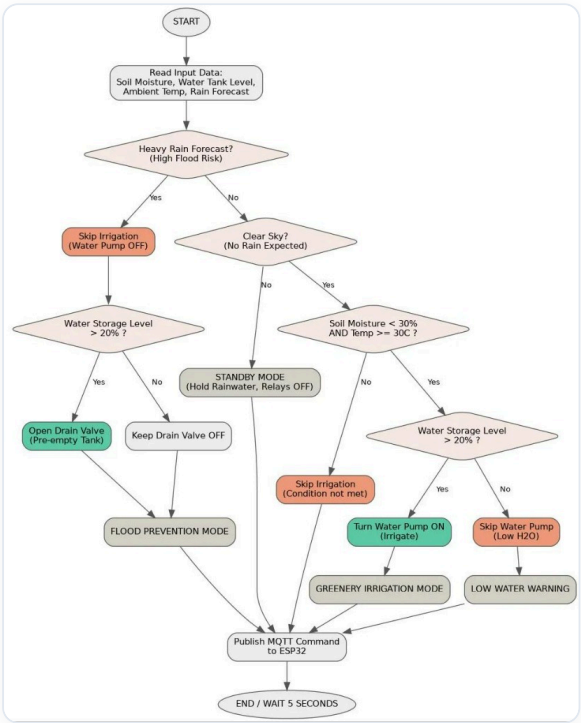
```
{
  "ts": 1786342072.4,
  "node": "ace-esp32-node1",
  "soil_moisture_pct": 27.4,
  "ambient_temp_c": 34.2,
  "tank_level_pct": 43.1
}
```

Command — Pi → ESP32, every 5 s

```
{
  "mode": "GREENERY_IRRIGATION",
  "pump": true,
  "drain": false,
  "reason": "moisture 27%, 34.2C, tank 43%",
  "ttl_s": 15.0
}
```

5-Second Decision Algorithm Flowchart

[System Decision Algorithm Flowchart]



Six Operating Modes

Operating Mode	Condition & Control Action
FLOOD_PREVENTION	P2 Pre-drain Pump active (empties tank to $\leq 10\%$ level)
GREENERY_IRRIGATION	P1 Water Pump 12 s pulse ON (~0.34 L micro-dosing)
IRRIGATION_SOAK	Waiting 10 min for soil absorption before re-evaluating
LOW_WATER_WARNING	Hot & dry, but tank water storage level $\leq 20\%$
STANDBY	Normal monitoring mode (holds harvested rainwater)
FAIL_SAFE	Sensor / API outage — all relay pumps shut OFF for safety

* **Priority is Strict:** Heavy rain forecast (flood prevention) takes absolute priority. Rain forecast arriving mid-irrigation instantly halts P1 and triggers P2 pre-drainage.

Engineering Solutions for Real Hardware

Six reliability solutions engineered to resolve sensor noise, deadband chatter, and communication dropouts in physical hardware.

Concern	As Specified	ACE Implemented Solution	Improvement / Why
Threshold Chatter	Irrigate below 30% moisture	Deadband hysteresis (dry <28%, recovered >35%)	Prevents rapid pump ON/OFF toggling near boundary readings
Over-irrigation Control	Pump ON while conditions hold	12 s pulse (~0.34 L) + 10 min soak hold	Prevents draining full tank while sensor slowly reacts
Pre-drain Stop	Open above 20%	Opens at 20%, closes at 10%, 300 s safety cap	Prevents pump dry-running and infinite drain loops
Bad Data / Outage	—	Stale >30 s or missing telemetry → FAIL_SAFE (all OFF)	A silent or failed sensor never misreads as "dry"

💡 Design Principle: All policy logic lives on the Raspberry Pi software (modifiable without reflashing), while the ESP32 maintains local hardware guardrails if connection drops.

Layer Stack & Hardware Component BOM

Crop Box Soil Profile Elevation (250mm Total)



Component	Qty	Specification & Role
ESP32 Dev Board	1	Edge node (Sensor readings & Relay control)
Raspberry Pi 4	1	Main Hub (MQTT Broker, decision engine, log DB)
Capacitive Moisture v1.2	1	ADC1 analog pin only (Corrosion resistant)
DHT22 Temp / Humidity	1	High-precision ambient temp (upgraded from DHT11)
HC-SR04 Ultrasonic	1	Non-contact water tank storage level sensor
5V Submersible DC Pump	2	P1 (Irrigation) + P2 (Pre-drainage)
2-Channel 5V Relay	1	ESP32 GPIO 26 / 27 pump switching

💡 **Design Upgrade: Solenoid Valve → Second Submersible Pump**
A 5 V solenoid valve requires ≈ 0.02 MPa to open. The 200 mm tank head provides 0.002 MPa — 10x too low. Replacing with a second submersible pump eliminates pressure requirements and ensures 100% reliable drainage.

Concept Balances at Scale: Water Balance Calculations

Catchment, storage capacity, and crop demand were verified against each other prior to building the physical prototype.

0.12 m²

Catchment Area
400 × 300 mm crop box

2.4 L

Captured per 20 mm rainfall
Fills 40% of 6 L tank

6 L

Tank Storage Capacity
200 × 150 × 200 mm

0.34 L

One Irrigation Pulse
12 s pulse (~1.7 L/min)

~15

Waterings from One Full Tank
Supports 7+ heatwave days

300 s

Maximum Drain Duration
Hard safety cap limit

Circular Resource Balance

A single 20 mm rainfall event harvests 2.4 L of rainwater into the tank, which subsequently powers roughly seven irrigation pulses during the following heatwave. The storm that threatens flooding is converted into the exact water that solves heat stress.

24-Hour Day Simulated Before Hardware Exists

python simulator.py --standalone --hours 24

```
time mode P1 P2 soil% tank% temp
00:00 STANDBY -- -- 45.0 50.0 25.6
09:20 GREENERY_IRRIGATION ON -- 30.4 47.6 30.0
09:21 IRRIGATION_SOAK -- -- 36.9 42.9 30.0
12:53 GREENERY_IRRIGATION ON -- 31.3 40.6 34.2
13:03 STANDBY -- -- 37.4 35.8 34.3
14:00 FLOOD_PREVENTION_DRAIN -- OPEN 34.8 32.4 34.8
14:00 FLOOD_PREVENTION_HOLD -- -- 34.8 8.1 34.8
16:03 FLOOD_PREVENTION_DRAIN -- OPEN 100.0 16.6 34.8
17:00 STANDBY -- -- 100.0 12.7 34.3

irrigated 0.85 L · drained 5.42 L · overflowed 0.00 L
```

09:20 — Heatwave Response

Temp reaches 30°C with soil at 27%. One 12 s pulse restores moisture 30% → 37%.

14:00 — Heavy Rain Forecast

Tank pre-drains 36% → 8% 1 hour before storm, clearing capacity for downpour.

16:03 — Storm Refills & Re-drains

Rainwater refills tank and pre-drain pump routes excess water. **Overflow: 0.00 L.**

15 / 15 Unit Tests PASS

Sensor fault, dry-run, stale telemetry, and forecast-interrupts-irrigation 100% verified.

Conventional Systems vs SDS Platform & Cooling Fog

Conventional Drainage vs SDS Platform Comparison

Feature	Conventional Municipal Drainage	ACE SDS Smart Platform
Primary Goal	Flood prevention only	Flood + Heatwave + Drought integrated control
Rainwater Processing	Store then discharge (wasted)	3-Stage Filter → Store → 100% Recycle (Irrigation & Mist)
Water Flow Mechanism	One-way gravity discharge	Bi-directional smart routing (P1 Irrigation & P2 Pre-drain)
Decision Timing	Reactive response after rain	Proactive 1-hour KMA forecast API poller
Application Scope	Large civil infrastructure only	Roofs, planter boxes, smart shade micro-zones

Smart Shade Cooling Fog Integration & Key Innovations

Rainwater Priority

Prioritizes filtered rainwater over municipal tap water, saving utility costs.

Multi-height Spray

Independent nozzle control for children/wheelchairs (1.0~1.2m) & adults (1.8m).

IoT Sensor Control

Real-time analysis of ambient temp/humidity to lower felt temp by 3~5°C.

Safety & Hygiene

Multi-layer fabric/gravel filtration ensuring mist hygiene and zero nozzle clogging.

Triple-Bottom-Line Expected Impact



1. Urban Disaster & Safety

- **Flood Risk Reduction:** Pre-draining frees capacity before heavy rainfall events.
- **Heat Island Mitigation:** Cooling fog mist reduces ground surface temperature.
- **Pedestrian Comfort:** Controls microclimate in urban rest spaces.



2. Smart Agriculture & Water

- **Automated Moisture:** Prevents crops from reaching wilting point.
- **Root Rot Prevention:** 1.2° incline plate ensures rapid gravity drainage.
- **Labor Efficiency:** Automates manual watering tasks for smart farming.



3. Economic & Social Value

- **Tap Water Savings:** Replaces municipal water with filtered rainwater.
- **Low Maintenance:** Self-cleaning 3-tier soil/filter/gravel media.
- **Scalable Architecture:** Modular edge node design scales across smart cities.

Smart Drain & Heatwave System Demonstration Video

TEAM ACE · SDS PLATFORM



1:1 WORKING PROTOTYPE LIVE DEMONSTRATION

"Let's Watch the Physical System Demonstration Video"

Demonstrating the 1:1 physical hardware prototype — featuring real-time water pump actions for P1 (Heatwave Irrigation) and P2 (Flood Pre-drainage), triggered by live weather API forecasts and sensor telemetry.



1. Heatwave Micro-Irrigation (P1 Pulse) Action



2. Pre-storm Smart Flood Drainage (P2) Action

SMART DRAIN & HEATWAVE RESPONSE SYSTEM

Thank you for your time. Ready for any questions.

Control Logic
100% Verified

Simulation Unit Tests
15 / 15 PASS

KMA Forecast API
Grid Conversion Validated

Question & Answer (Q&A)

Team ACE · 2026